

0907 U1 quantitative data FRQ

1. A country's transportation department wants to conduct a survey to collect data from adults, ages 18 and older, in that country. The survey will be focused on commuting to work, including commute time in minutes, method of transportation used, and other information.

Country A is divided into many regions, and each region will be asked to send out a survey and collect the data. The transportation department will allow each region to use any method it prefers to survey the adults.

Three different regions will use the following sampling methods:

Region X: Each adult who lives in the region will receive a survey and will be required to complete the survey by mail. Adults who do not complete the survey will be sent reminders to do so until they have completed the survey.

Region Y: The region is divided into six areas that are similar to each other. Two of the six areas will be randomly selected, and each adult who lives in those two areas will receive a survey and be asked to complete the survey by mail. The table gives the results of the data collection for the two areas in Region Y for the question "What is your average daily commute time to work in minutes?"

Region Z: Each adult who lives in the region and has responded to a previous official survey will receive a survey and be asked to complete it by mail.

Summary Statistics

Area	$\bar{x}$	n	Min	Q1	Median (Q2)	Q3	Max
One	55	2,887	7	35	53	74	101
Five	42	4,502	5	24	32	68	83

Respond to parts A, B, C, D, and E, and any subparts.

Part D

Use comparisons of the summary statistics in the table to **describe** the most likely shape for the distribution of commute time in Area Five.

Part D Point 1

Write the response in complete sentences. Use appropriate psychological terminology.

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution

- The mean (42) is larger than the median (32), which suggests that the distribution of commute time for Area Five is likely right-skewed.



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0907 U1 quantitative data FRQ

The student response accurately **describes numerical representations of data.**

Responses that earn this point:

- The response compares the median of 32 to the mean of 42 for Area Five and describes that the distribution is likely somewhat right-skewed.
- The response indicates a comparison using the five-number summary and describes that the distribution is likely somewhat right-skewed.

Examples that earn this point:

- **For Area Five, the distance from Q1 to the median is smaller than the distance from the median to Q3, indicating that this distribution is likely right-skewed.**

0907 U1 quantitative data FRQ

2. Peyton, a researcher at a polling company, wants to conduct a study to investigate if, in a certain country, voting preferences differ in three areas (Urban, Suburban, and Rural) where registered voters live. Each area has over 100,000 registered voters. Each selected registered voter will be asked the following survey question: “Do you plan on voting in the upcoming election?” with answer choices of Yes, No, or Not Sure. The researcher is considering choosing one of the following sampling method plans.

Plan X: From each of the three areas (Urban, Suburban, and Rural), 300 registered voters will be randomly selected.

Plan Y: A simple random sample of 900 registered voters will be taken from the most populated area, Suburban.

The table gives the results of using Plan X to collect the data.

Data from Plan X Sampling Method

	Urban	Suburban	Rural
Yes	180	120	100
No	30	115	90
Not Sure	90	65	110

Respond to parts A, B, C, D, and E, and any subparts.

Part C

Peyton wants to construct a graphical representation using the data in the table.

- Identify** a graphical representation that is appropriate to use to display the data.
- Explain** why the graphical representation from part C (i) is appropriate to use.

Part C Point 1

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution:

A segmented bar graph or mosaic plot



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The student response accurately **identifies** graphical representations of the data.

Responses that earn this point:

0907 U1 quantitative data FRQ

- The response identifies a segmented bar graph.
- The response identifies a side-by-side bar chart (bar graph).
- The response identifies a mosaic plot.

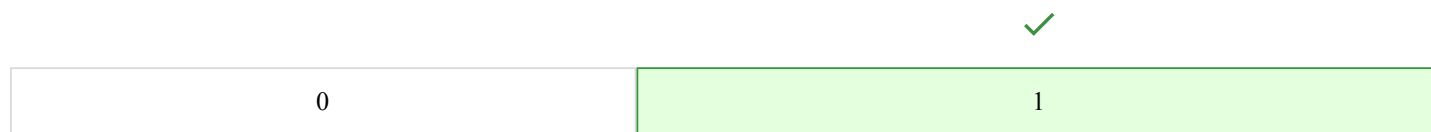
Part C Point 2

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution:

The graph displays each of the three answer choices by area since both variables are categorical data and show the relative frequencies of each answer choice.



The student response explains why graphical representations are appropriate to display the data.

Responses that earn this point:

- The response accurately explains that the graph displays each of the three answer choices by area since both variables are categorical data and show the relative frequencies of each answer choice.
- The response accurately explains that the graph displays the relative frequencies for the categories of the areas by each answer choice.
- A response that incorrectly identifies the graphical representation but gives a correct reason earns Part C Point 6 (but does not earn Part C Point 5).

0907 U1 quantitative data FRQ

A city’s public health department recently started a “Hydration and Health” campaign to encourage adults to drink enough water each day. In the city, doctors have reported that many adults substitute caffeinated beverages, such as coffee and tea, for water. They are worried this might keep adults from staying properly hydrated.

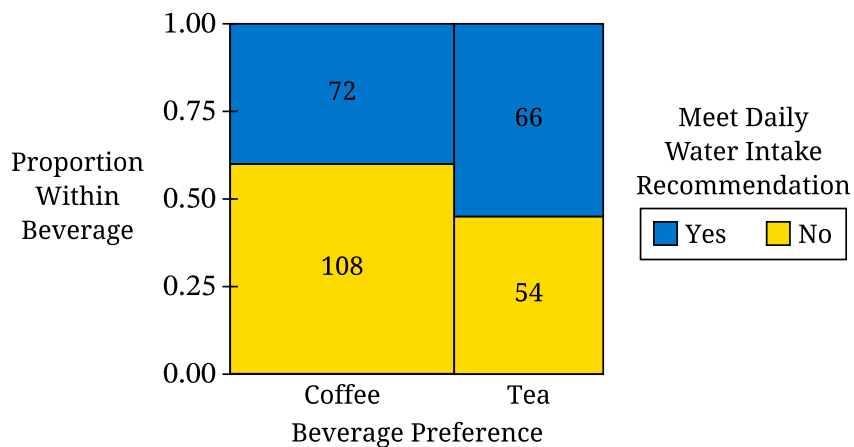
To look into this issue, the department surveyed a random sample of 300 adults from across the city. Each person filled out a short survey about their beverage preference and whether they meet the daily water intake recommendation. The following variables are used to create a table and mosaic plot.

- Beverage Preference: Coffee or Tea
- Meets Water Intake Recommendation: Yes or No

The department plans to use the survey results to help create educational materials for the campaign. City officials also hope the information will show whether new ideas—like adding free water refill stations—might help adults who are not drinking enough water. The survey results from the 300 residents are shown in the table.

Beverage Preference and Meets Water Intake Recommendation

		Meet Daily Water Intake Recommendation		
		Yes	No	Total
Beverage Preference	Coffee	72	108	180
	Tea	66	54	120
	Total	138	162	300

Beverage Preference and Meets Water Intake Recommendation

0907 U1 quantitative data FRQ

3. Respond to parts A, B, C, D, and E, and any subparts.

Part A

- i. **Classify** the variables in the study.
- ii. **Identify** the population in the study.

Part A Point 1

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution:

- Both variables, Beverage Preference and Meets Water Intake Recommendations, are categorical variables in the study.



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The student response accurately **classifies** the relevant information.

Responses that earn this point:

- Both variables are categorical.

Examples that earn this point:

- both variables are qualitative.
- both variables, Beverage Preference and Meets Water Intake Recommendations are qualitative variables

Part A Point 2

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution:

- The population in the study is all adults from across the city.



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The student response accurately **identifies** the relevant information.

Responses that earn this point:

0907 U1 quantitative data FRQ

- The response accurately identifies the population in the study as all adults from across the city.

Examples that earn this point:

- all adults who live in the city

0907 U1 quantitative data FRQ

4. Respond to parts A, B, C, D, and E, and any subparts.

Part C

- i. **Identify** an appropriate graphical representation, that is different from the mosaic plot, for displaying the data shown in the table.
- ii. **Explain** why there is an advantage to using the mosaic plot instead of the graphical representation identified in part C (i).

Part C Point 1

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution:

- A side-by-side bar chart



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The student response accurately **identifies** graphical representations of the data.

Responses that earn this point:

- The response identifies a side-by-side bar chart.
- The response identifies a segmented bar chart.

Examples that earn this point:

- a side-by-side bar graph
- a segmented bar graph

Part C Point 2

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution:

- There is an advantage to using a mosaic plot compared to a side-by-side bar chart because the mosaic plot displays the difference in sample sizes by having a wider or more narrow width on the horizontal axis for the beverage preference (coffee and tea) which a side-by-side bar chart does not.

0907 U1 quantitative data FRQ



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The student response **explains** why graphical representations are appropriate to display the data.

Responses that earn this point:

- The response accurately explains that the advantage of using a mosaic plot compared to a side-by-side bar chart is that the mosaic plot displays the fact that the sample size of coffee drinkers was larger than that of the tea drinkers and demonstrates the proportion of each group that met daily water intake. However, the side-by-side bar charts just demonstrates the proportion of each group that met daily water intake.
- A response that incorrectly identifies the graphical representation but gives a correct reason earns Part C Point 6 (but does not earn Part C Point 5).

Examples that earn this point:

- There is an advantage to using a mosaic plot compared to a side-by-side bar graph because the mosaic plot displays the difference in sample sizes for each beverage preference (coffee and tea).
- There is an advantage to using a mosaic plot compared to a segmented bar chart because the mosaic plot displays the difference in sample sizes for each beverage preference (coffee and tea).
- There is an advantage to using a mosaic plot compared to a segmented bar graph because the mosaic plot displays the difference in sample sizes by having a wider or more narrow width on the horizontal axis for each beverage preference (coffee and tea).

0907 U1 quantitative data FRQ

5. Andrea, a cybersecurity specialist at a bank, was concerned about the security of the bank's website. She was asked to compare two encryption processes where information is scrambled and cannot be easily read methods that could be used to protect the website's banking information. Her department was concerned that the older encryption version, DES, may be weaker than the newer encryption version, AES. One measure of how well an encryption method works is session duration, how long the information is accessed in seconds. A very long session duration may indicate that there has been unauthorized access to the data, which means that the encryption method needs to be more secure. Her department wanted her to study the two encryption methods.

Therefore, Andrea conducted a study of the two types of encryption methods. Andrea randomly collected this information from bank customers who used the website over the previous six months. For each session of the sampled customers, Andrea collected the session duration, in seconds, and the type of encryption method used. The table displays the summary statistics for her study.

Summary Statistics

Encryption Method	n	$\bar{x}$	s	Min	Q1	Median (Q2)	Q3	Max
DES	2,865	782	768	1	226	543	1,107	6,950
AES	4,706	794	803	1	233	794	1,091	7,190

Respond to parts A, B, C, D, and E, and any subparts.

Part C

Use the table to respond to the following.

- Compare** the distance from Q1 to the median and the distance from the median to Q3 for the DES encryption method.
- Compare** the distance from Q1 to the median and the distance from the median to Q3 for the AES encryption method.
- Explain** how these comparisons can be used to select which measure of variability—standard deviation, range, or IQR—would be the best representation for both of the encryption methods.

Part C Point 1

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

0907 U1 quantitative data FRQ

Sample Solution

- For the DES encryption method, the distance from Q1 to the median (317) is less than the distance from the median to Q3 (564).



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The student response accurately **compares** numerical representations of data.

Responses that earn this point:

- The response indicates that the distance from Q1 to the median is less than the distance from the median to Q3.
- The response indicates that the distance from Q1 to the median is less than the distance from the median to Q3 but gives incorrect values. However, the work shown indicates that this is a minor mathematical error.

Examples that earn this point:

- For the DES encryption method, the distance from Q1 to the median ($543 - 226 = 217$) is less than the distance from the median to Q3 ($1,107 - 543 = 564$). (Work is shown and 217 is treated as a minor mathematical error.)

Part C Point 2

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution

- For the AES encryption method, the distance from Q1 to the median (561) is greater than the distance from the median to Q3 (297).



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The student response accurately **compares** numerical representations of data.

Responses that earn this point:

- The response indicates that the distance from Q1 to the median is greater than the distance from the median to Q3.
- The response indicates that the distance from Q1 to the median is greater than the distance from the median to Q3 but gives the incorrect values. However, the work shown indicates that this is a minor mathematical error.

Examples that earn this point:

- For the AES encryption method, the distance from Q1 to the median ($794 - 233 = 461$) is larger than the distance from the median to Q3 ($1,091 - 794 = 297$). (Work is shown and 461 is treated as a minor mathematical error.)

Part C Point 3

0907 U1 quantitative data FRQ

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution

- The best measure of variability for both methods is the IQR because the distributions are not symmetric.



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The student response accurately **explains** numerical representations of data.

Responses that earn this point:

- The response uses the comparison of the distances from Q1 to the median and the median to Q3 to indicate that the IQR is the best measure of variability because it is evident that neither method is symmetric.
- The response shows evidence that neither method is symmetric, which will affect the selection of the measure of variability.

Examples that earn this point:

- Because the distance from Q1 to the median and the distance from the median to Q3 are very different within each distribution, the distributions are likely skewed. Therefore, the IQR is the best measure of variability for both methods.